

Spike: 5**Title: Portfolio 1 Task 13: Spike - Tactical Steering (Hide!)****Author:** Edward Herrod, 104541862**Goals / deliverables:**

The main aim of this spike is to demonstrate a working simulation of a hunter and a prey agent navigating a toroidal world with multiple circular objects ("rocks"). Agents are visually distinct from one another through colour (hunter is red and prey is yellow) and have slightly different masses so that chases are asymmetric. Hitting the "I" key renders indicators which show where the prey agent is considering hiding as green diamonds (the one it is pursuing is red). The prey agent automatically chooses the closest hiding location and steers towards it whilst avoiding obstacles.

Technologies, Tools, and Resources used:

List of information needed by someone trying to reproduce this work:

- Visual Studio Code
- Python 3.x
- Pyglet 2.1.x
- Git
- Github (for installation of code)

Tasks undertaken:

- Download and install Visual Studio Code
- Download and install the latest version of Python
- Install the latest version of Pyglet (pip install --upgrade --user piglet)
- Import the lab containing the base code for this spike (Task 11)
- Create Rock objects in World.py
- Update agent code to include pursuit and fleeing
- Update agent code to include obstacle avoidance, collision and line-of-sight for the hunter
- Include all required debugging information

What we found out:

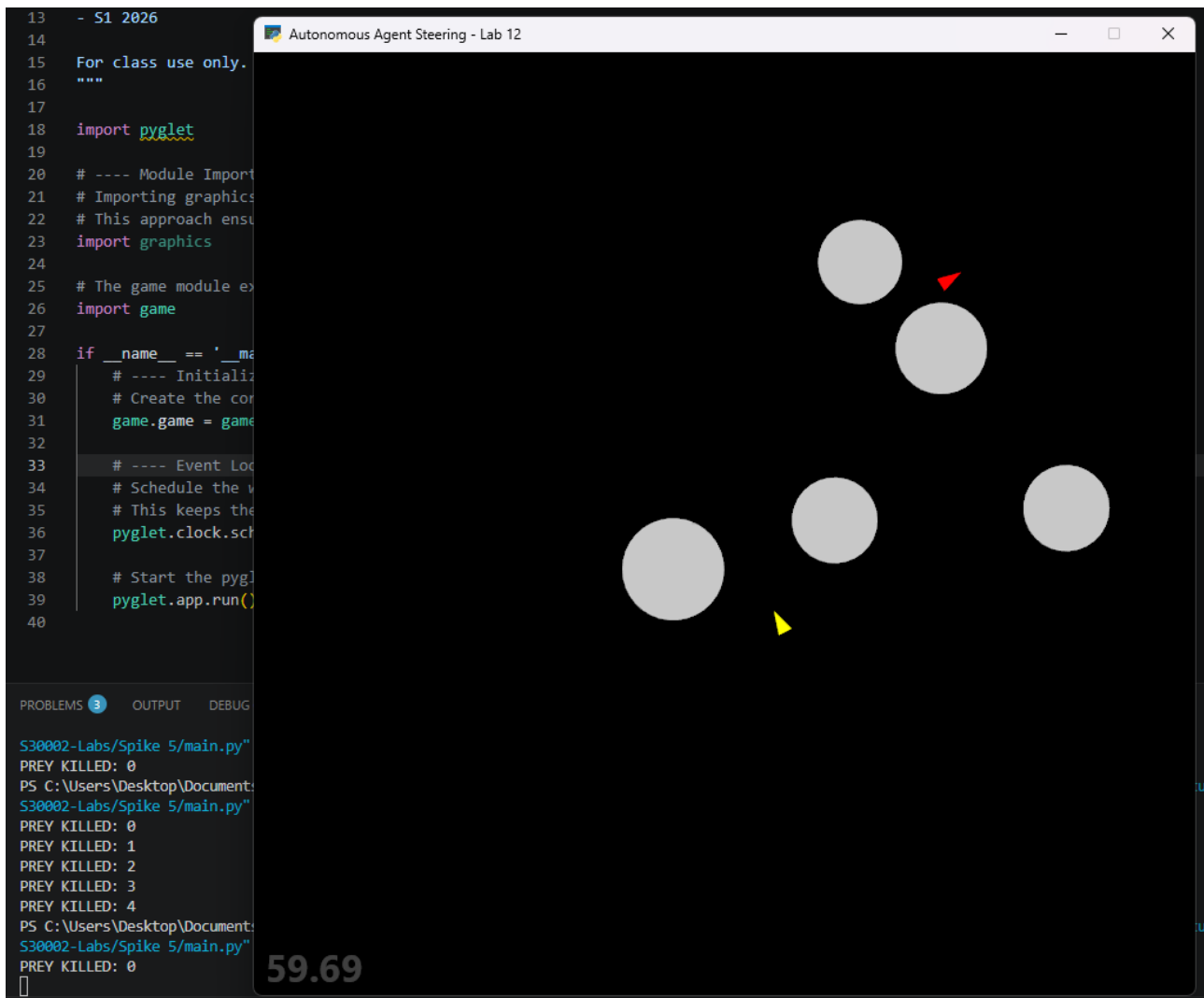


Figure 1: The prey agent is hiding behind a rock opposite of the hunter agent.

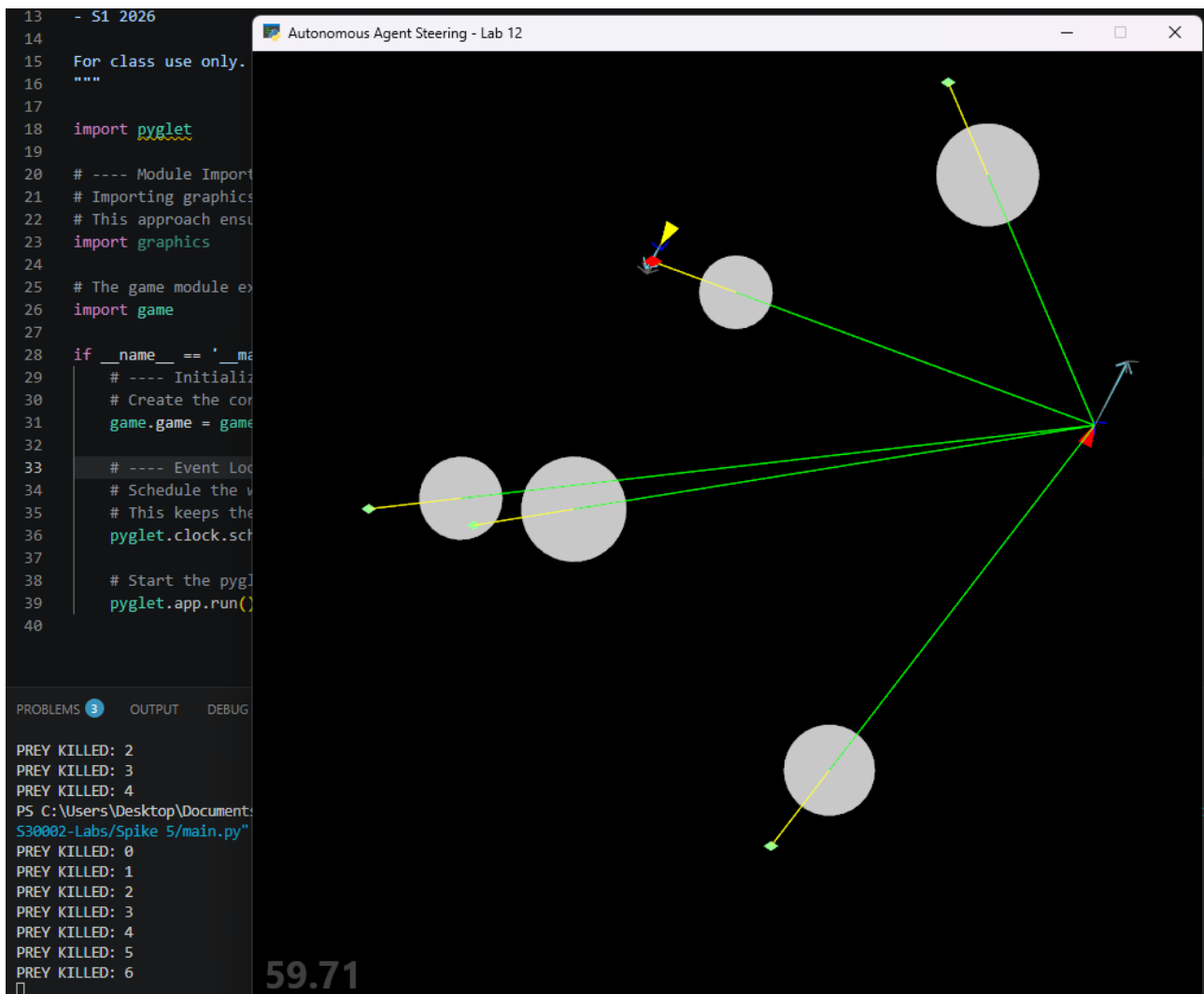


Figure 2: Debug information displays all the possible hiding points and their relation to the hunter's position, demonstrating that they are directly opposite the rocks from it. The hiding spot that the prey has chosen is highlighted red. The rocks have been randomised by pressing "C" and the terminal displays how many times the prey agent has been killed and respawned.